

DeadEye handbook

pre-publication guide, research plan, pre-registration, peer testing, budget, limitations, references audit

8 October 2026

- [Before you publish](#)
- [Research plan](#)
- [Compute budget](#)
- [Known limitations](#)
- [References audit](#)

Before you publish

This page is written for the person who will run the study and submit the paper. It says plainly what exists today, what is still missing, and the order in which to do the remaining work.

What is done

- The benchmark software is complete and tested: seven decision tasks with known optimal policies, four model backends (local Hugging Face weights, OpenAI-compatible servers, decision-model APIs, and a mock for tests), six conversion methods, a resumable runner that logs every decision, statistics (confidence intervals, paired tests, scale slopes, calibration), and a report generator that writes every table and figure the paper uses. `pytest` and `deadeye` smoke pass.
- The paper is drafted in full: abstract, introduction, related work, benchmark, setup, results structure, analysis plan, limitations, conclusion and appendix (every prompt verbatim), with a verified Harvard bibliography of 124 entries.
- The research plan, pre-registration, compute budget, peer-testing guide and known limitations are written.

What is not done

- **No real model has been evaluated.** The sandbox in which this was built had no GPU and no access to Hugging Face, so every experiment ran only on a mock model and a tiny randomly initialised model, which prove the pipeline works and nothing else.
- **Every number in the paper's results is a placeholder.** Placeholders are the blue `\result{...}` marks (127 of them) and the red `\todo{...}` notes (23). The draft PDF prints them in colour so none can be missed.
- **The pre-registration is not frozen** (`docs/preregistration.md` still has blanks for the pilot's standard deviations), and nothing has been tagged.
- **Two systems need a small adapter** before they can be scored: Laya's package and Perplexity's Decisions API use their own request schema; Kev and Clef speak the protocol already implemented.
- **Two small source checks** remain: the TechCrunch bylines in the bibliography, and the Qwen3.5 repository ids in `configs/decision_models.yaml`, which must be confirmed on the Hugging Face hub.

What to do, in order

Each step says what you need, what to run, how long it takes, and what you should see.

Step 1. Get a machine and install (half a day) You need a Linux machine with an NVIDIA GPU. One 24 GB card (for example an RTX 4090 or L4) handles every model up to 14B parameters; the 27B to 72B rows need an 80 GB card (A100 or H100) or 4-bit loading. Cloud rental is fine. Then:

```
git clone https://github.com/CaptiveON/DeadEye && cd DeadEye
python -m venv .venv && source .venv/bin/activate
pip install torch --index-url https://download.pytorch.org/whl/cu124 # pick the CUDA build for your driver
pip install -e ".[hf,quant,dev]" tabulate
pytest -q && deadeye smoke
huggingface-cli login # needed for gated repositories (Llama, Gemma)
```

You should see all tests pass and a smoke-test report in a temporary folder.

Step 2. Run the pilot (a few hours on a GPU; a day on a laptop CPU)

```
deadeye run configs/pilot_cpu.yaml
deadeye report results/pilot_cpu --out report/pilot_cpu
```

Open `report/pilot_cpu/index.html`. Check three things: the random row scores 0 and oracle scores 1 on every task; the illegal-action rate under `prompt_generate` is below about 20% for the instruct models (if it is much higher, read the raw outputs in `steps.jsonl` and fix the prompt wording in `src/deadeye/policies/prompts.py`); and the probe's training accuracy in `summary.json` is well above chance. Prompts may be edited only during this step.

Step 3. Freeze the design (one hour) Fill in the per-task standard deviations in section 5 of `docs/preregistration.md` from `report/pilot_cpu/episodes.csv` (the standard deviation of the paired differences of norm between two methods on the same seeds), then:

```
git add -A && git commit -m "Freeze pre-registration" && git tag prereg-v1 && git push --tags
```

Optionally register the same document on OSF (osf.io) for a timestamp outside your repository.

Step 4. Run the main sweep (about 150 GPU-hours on one 80 GB card; 2 to 4 days on 2 to 4 cards)

```
deadeye estimate configs/sweep_gpu.yaml --seconds-per-decision 0.3 # check the budget first
scripts/run_sweep.sh configs/sweep_gpu.yaml 0 1 2 3 # one process per GPU id
deadeye run configs/sweep_controls.yaml # Pythia and OLMO 2 (about 60 GPU-hours)
```

Runs are resumable: if a machine stops, run the same command again and finished cells are skipped. Watch `results/sweep_gpu/run_manifest.json` for the cell list and `logs_*.txt` for errors.

Step 5. Run the ablation blocks (about the same again; each block can run alone)

```
for b in quantisation prompting thinking adaptation tasks; do deadeye run configs/ablations/$b.yaml; done
```

Step 6. Run the decision-model comparison (one to two days, mostly waiting on servers)

- Kev: install from github.com/jaredpalmer/kev and start one server per size on ports 8009, 8010, 8011 as written in `configs/decision_models.yaml`.
- Clef: set `CLOUDFLARE_ACCOUNT_ID` and `CLOUDFLARE_API_TOKEN`; Workers AI bills per request.
- Jev: apply for early access at typesafe.ai and set `TYPESAFE_API_KEY`; if access is not granted, leave the row out and say so in the paper.
- Laya and Perplexity: write the adapter (copy `src/deadeye/models/decision_api.py`, change the request and response field names to match their documentation, register it in `src/deadeye/models/registry.py`).
- SemIf: run its server, or rely on the `score_letter` row, which is the same computation.
- Confirm the three Qwen3.5 base repository ids on the hub and correct them in the config if needed.

```
deadeye run configs/decision_models.yaml
```

Step 7. Generate every table and figure (minutes)

```
deadeye report results/sweep_gpu results/sweep_controls results/ablations results/decision_models \
--out report/paper --paper-dir paper
```

This writes `paper/tables/*.tex` and `paper/figures/*.pdf`; the paper includes them automatically. `report/paper/cells.csv`, `slopes.csv`, `pairwise_methods.csv` and `system_comparison.csv` hold every number.

Step 8. Fill the results (two to three days of careful work) Work through `paper/sections/results.tex`, `analysis.tex`, `abstract.tex`, `introduction.tex` and `conclusion.tex`. Replace each `\result{...}` with the measured value and its confidence interval, taken from the CSV files, and delete each `\todo{...}` once it is handled. For the paired tests, run one `deadeye compare` call per hypothesis family (the pre-registration lists them) and quote the Holm-adjusted p-values. Then check the pre-registered criteria one by one and write "supported" or "falsified"; a falsified hypothesis is a result, not a failure.

```
grep -c '\\result{' paper/sections/*.tex # must reach 0
grep -c '\\todo{' paper/sections/*.tex # must reach 0
python scripts/check_tex.py # braces, environments, citation keys
```

Step 9. Final source checks (one hour) Open the TechCrunch articles and add the journalists' names to `paper/refs.bib`; open each model card cited in the decision-model table and confirm the sizes and licences still match; re-run `python scripts/check_tex.py`. **Step 10. Build the PDF and submit** With a TeX installation: `cd paper && make`. Without one: `python scripts/build_draft_pdf.py` builds the same content through pandoc and Chromium. Archive the `results/` folder and the report with a DOI (Zenodo), tag the repository v1.0, and submit (arXiv first, then the venue). The reproducibility checklist in the appendix is written for this moment.

Honest expectations

The design was reviewed adversarially during construction and every oracle was checked, but no study survives contact with real models unchanged. Expect to find a prompt that a model family misreads, a checkpoint that fails to load, or a task whose variance is higher than planned. All of these are handled by the plan: prompts change only before the tag, failed cells are listed in the appendix, and the power section reports the smallest effect the design could detect instead of claiming a null.

Research plan

The question we want to answer is simple to state and surprisingly open: when you take an open-weight language model and turn it into something that makes decisions, what actually determines how good those decisions are? Everyone assumes bigger is better. But "turning a model into a decision-maker" can mean prompting it, scoring its likelihoods, training a small head on its hidden states, or fine-tuning it on demonstrations, and these use the weights in very different ways. Our bet is that the conversion method and the shape of the task matter at least as much as the parameter count, and that a small model handled well will often match a much larger one handled naively. The experiments below are designed to find out whether that bet holds, and where it fails.

Three things come out of the project. The first is the deadeye harness in this repository: a benchmark that anyone can install with pip, run on a laptop for a pilot or on a GPU box for the full sweep, and that produces a leaderboard from its own logs. The second is a paper (skeleton in `paper/`) in which every table and figure is generated by `deadeye report`, so nothing in it is typed by hand. The third is a pre-registration (`docs/preregistration.md`) that pins down the hypotheses and the analysis before the main sweep runs, so that we cannot quietly move the goalposts afterwards.

What follows is the whole route: where to start, what each phase produces, how to test it, and what "finished" looks like. Every command in this document runs today.

1. Framing

1.1 What "converting a weighted model into a decision model" means here A decision model is a policy: it maps a state or observation to an action. A language model is not one; it is a distribution over text. Bridging the two is where most of the design choices hide, and it is usually done one way, without comment, in papers that then draw conclusions about "the model". We treat the conversion method as an experimental factor in its own right, with five levels:

Method (<code>methods[].name</code>)	What is used from the model	Training data	Where it runs
<code>prompt_free</code>	the unconverted assistant: it answers the situation in its own words and an action is extracted from the reply if one can be; the reference point for "before conversion" latency, tokens and usable-decision rate	none	any backend
<code>prompt_generate</code>	free-form generation, parsed into an action (zero-shot, few-shot, chain-of-thought variants)	none	any backend
<code>prompt_score</code>	exact log-likelihood of each legal action as a continuation (lm-eval-harness style)	none	hf; first-token approximation on <code>openai</code>
<code>probe</code>	hidden state of the prompt at one layer, with a logistic-regression head	oracle-labelled states from training seeds	hf
<code>lora_sft</code>	LoRA behaviour cloning on oracle (observation, action) pairs, then scoring	same	hf
<code>feature_probe</code>	no language model: the same logistic head on the environment's raw numeric features	same	anywhere

`prompt_free` is the "before" condition every other method is measured against. `feature_probe` is the control that answers "does the language model's representation add anything over the raw features?". `random` and `oracle` anchor the score scale; `ucb1` is the classical algorithmic reference on the bandit.

1.2 Tasks All environments are procedurally generated with a seed and have a known oracle, and every rendering contains the full observable state. For the two bandit tasks the oracle is clairvoyant: it knows the hidden arm means or weights, which no policy can read off the text. There it serves as the upper anchor and as the label source for cloning, while UCB1 marks what an algorithm that sees only the history can achieve; the cloned policies learn a history-based approximation of "pull the best arm". They span the decision structures that matter in practice:

Environment	Decision structure	Oracle	Headline metric
<code>bandit</code>	exploration vs exploitation, no state	best true arm	pseudo-regret
<code>contextual_bandit</code>	in-context learning of an abstract rule from (context, action, reward) history	argmax of hidden logistic model	regret
<code>loan</code>	one-shot decisions from semantically meaningful features (tabular, knowledge-driven); OOD switches	approve iff true expected value > 0	profit, regret
<code>gridworld</code>	deterministic multi-step planning	BFS shortest path	success, excess steps
<code>tictactoe</code>	adversarial, perfect information	minimax	win rate
<code>blackjack</code>	risk under uncertainty	basic strategy (hit/stand)	expected return
<code>support</code>	applying a written company policy to a customer's message ("I received the package damaged. I need a refund today.")	the policy's rules	correct-decision rate, payout waste

Every episode return is normalised so that the random policy scores 0 and the oracle scores 1 on that environment; a normalised score is comparable across tasks and the cross-task mean is a single headline number per (model, method).

1.3 Factors

Factor	Levels in the configs	Hypothesis
Parameter count (within one family / recipe)	Qwen2.5 0.5B to 72B (primary: one recipe, base + instruct at every size); Qwen3 0.6B to 32B; SmoLLM2 135M to 1.7B; Gemma 3 270M to 27B; Pythia 70M to 12B and OLMo 2 1B to 32B as fully open controls (<code>configs/sweep_controls.yaml</code>)	H1
Instruction tuning	base vs instruct checkpoint of the same size (Qwen2.5 at all seven sizes; Qwen3 only up to 14B, which has no larger base checkpoints)	H2
Conversion method	the five methods above	H3, H4
Quantisation	bf16, int8, int4	H5
Observation / history format	<code>bandit history_format: summary</code> vs <code>raw</code> ; <code>history_window</code>	H6
Reasoning budget	Qwen3 thinking on/off; <code>cot</code> ; <code>max_new_tokens</code> 16 / 256 / 2048	H7
Prompt details	<code>action_format: name</code> vs <code>letter</code> ; <code>few_shot</code> 0/3; <code>temperature</code> 0/0.7; <code>length_norm</code>	secondary
Data budget for adaptation	<code>probe</code> / LoRA with 10, 40, 100 training episodes	H3
Task difficulty	arms 5 vs 10, grid 6 vs 9, opponent random vs minimax, loan shift none / covariate / sign-flip	H1, H3

Notes from the verified literature (`docs/literature_review.md`): the closest prior designs are Jiwatode et al. (2026), who cross Qwen3 sizes with thinking on/off on grid games, and Schmied et al. (ICLR 2026), who find 2B-27B models leave up to 55% of bandit arms untried and that correct rationales often do not translate into the chosen action. H3 has partial prior answers (SFT memorises while RL generalises; a post-trained 3B beating a 72B on Frozen Lake; but diverse-task fine-tuning transfers to unseen tasks), so our contribution on H3 is the controlled, within-family, pre-registered measurement rather than the direction of the effect. Newer Qwen ladders (Qwen3.5 0.8B-397B, Qwen3.6/3.8) exist and can replace Qwen3 once their repo ids are verified in `docs/model_ladders.md`. Gemma's large embedding share argues for plotting non-embedding parameters where published (`non_embedding_params` in the catalogue).

1.4 Pre-registered hypotheses (details and falsification criteria in `docs/preregistration.md`)

- H1 (scale).** Under `prompt_generate`, normalised score increases approximately log-linearly with parameters within a family, with a task-dependent slope: steep for planning and adversarial tasks (gridworld, tictactoe), shallow for one-shot contextual decisions (loan).
- H2 (instruction tuning).** At equal size, instruct checkpoints beat base checkpoints under prompting; the gap shrinks to non-significance under `probe` and `lora_sft`.

- **H3 (adaptation vs scale).** A lora_sft or probe model of at most 1.7B parameters matches or exceeds a zero-shot prompt_generate model ten times larger on in-distribution instances, but not on the out-of-distribution loan variants, where the generalisation gap grows with size.
- **H4 (scoring vs generation).** prompt_score eliminates format failures and closes a substantial fraction (we predict more than half) of the gap between the smallest and largest model under prompt_generate.
- **H5 (quantisation).** int4 costs less than 0.05 normalised score at 7B and above and more than 0.10 below 2B.
- **H6 (exploration).** On the bandit, no model at any scale reaches the UCB1 reference under prompting; summary history beats raw history for small models.
- **H7 (reasoning).** Thinking mode / chain-of-thought improves gridworld and tictactoe for models of 4B and above, does not improve or hurts models below 2B, and always loses on latency-normalised efficiency.

2. Phases: where to start, what to do, what each phase produces

Phase 0: harness and sanity (done) This is the part that already exists: six environments with oracles, three backends (hf, openai, mock), five methods, a resumable runner that logs every decision, the statistics (bootstrap confidence intervals, interquartile means, paired permutation tests with Holm correction), and a report generator that writes Markdown and LaTeX tables, figures and an HTML page. The exit criteria are met if the following all pass, and they do:

```
pip install -e ".[hf,dev]" tabulate
pytest -q # unit + integration tests, tiny random model, no downloads
deadeye smoke # full pipeline end-to-end in ~30 s, writes an HTML report
deadeye list-envs; deadeye list-methods
deadeye show-prompt gridworld --seed 3 --cot --few-shot 2 # see exactly what a model sees
```

Phase 1: CPU pilot (week 1-2) Goal: validate the pipeline on real weights, fix prompt bugs, and get first effect sizes to size the main sweep. Nine models up to 1.7B (SmolLM2, Qwen2.5, Qwen3), four methods, shortened horizons.

```
deadeye estimate configs/pilot_cpu.yaml --seconds-per-decision 0.5
deadeye run configs/pilot_cpu.yaml --only-model HuggingFaceTB/SmolLM2-135M-Instruct # ~20 min
deadeye run configs/pilot_cpu.yaml # all, resumable
deadeye report results/pilot_cpu --out report/pilot_cpu
```

One rule governs the pilot: prompts may be edited here and nowhere else, and only against the pilot seeds. When the pilot is done, tag the repository (git tag prereg-v1) and freeze docs/preregistration.md. From that tag on, every run is confirmatory, and any change to the design goes into the deviations table with a date and a reason.

Outputs: report/pilot_cpu/index.html, a first scale curve, an estimate of the per-episode standard deviation of normalised score on each task (used for the sample-size calculation), and a list of format failures from steps.jsonl (search for "legal": false).

Phase 2: main sweep on GPU (week 3-5) The within-family ladders with all methods, 100 episodes per cell (300 for tic-tac-toe, whose outcomes are -1/0/+1; blackjack episodes are 20 hands each), 100 training episodes for probes and LoRA. configs/sweep_gpu.yaml has 33 model entries (Qwen2.5 base and instruct at all seven sizes, Qwen3 base up to 14B and post-trained up to 32B, SmolLM2, Gemma 3) and 6 methods; configs/sweep_controls.yaml adds the fully open Pythia and OLMo 2 ladders. Cells are resumable, so a sweep can be split across machines by --only-model (scripts/run_sweep.sh does this).

```
deadeye estimate configs/sweep_gpu.yaml --seconds-per-decision 0.3
CUDA_VISIBLE_DEVICES=0 deadeye run configs/sweep_gpu.yaml --only-model Qwen/Qwen2.5-7B-Instruct
deadeye report results/sweep_gpu --out report/sweep_gpu --paper-dir paper
```

Hardware: everything up to 14B runs in bf16 on a 24 GB GPU; 27B-72B use quantization: int4 (bitsandbytes, pip install -e ".[quant]"), which needs about 20-45 GB. Mark quantised rows as such in the paper; the quantisation ablation in Phase 3 measures what int4 costs at sizes where both fit.

Phase 3: ablations (week 5-7) The five files in configs/ablations/ isolate one factor each on three sizes of the Qwen2.5 ladder (1.5B, 7B, 14B), plus the Qwen3 ladder for thinking: quantisation (H5), prompting (demonstrations, chain of thought, temperature, action encoding, history window, length normalisation), thinking (H7), adaptation (data budget and probe layer, H3) and tasks (observation format, action-space size, adversary strength, planning depth, and the out-of-distribution loan variants, which train the adaptation methods in distribution through train_params). Each block is a few hundred cells; run them one at a time:

```
deadeye run configs/ablations/quantisation.yaml
deadeye run configs/ablations/tasks.yaml --only-env loan_sign_flip
deadeye report results/ablations --out report/ablations # aggregates all blocks
```

Phase 4: analysis and writing (week 7-9) deadeye report produces every table and figure, including the scale-slope fits (slopes.csv) behind H1 and H4b; deadeye compare runs the paired permutation tests behind each claim, one hypothesis family per call so that the Holm correction is applied within the family:

```
deadeye compare results/sweep_gpu \
  --pair "loan:Qwen_Qwen2.5-1.5B-Instruct/lora_sft vs loan:Qwen_Qwen2.5-14B-Instruct/prompt_generate" \
  --pair "gridworld:Qwen_Qwen2.5-1.5B-Instruct/lora_sft vs gridworld:Qwen_Qwen2.5-14B-Instruct/prompt_generate"
``` Fill `paper/sections/*.tex` from the report's `tables/` and `figures/`. The
literature review in `docs/literature_review.md` and the bibliography in `paper/refs.bib` are
pre-verified.

Phase 5: release (week 9-10)

Tag `v1.0`, build the Docker image, archive the results directory and the report with a DOI
(Zenodo), publish the leaderboard HTML, submit the paper (targets: NeurIPS Datasets & Benchmarks,
TMLR, COLM, or an agents / small-models workshop), post the preprint.
```

```

3. How to test: validity, statistics, and the checks that must pass

3.1 Harness validity (automatic)

- Oracle regret is 0 and the oracle's normalised score is 1 on every environment; the random policy's is 0 (`tests/test_envs.py`, `tests/test_metrics_report.py`).
- Each seed reproduces exactly the same instance and the same pre-drawn noise, so two policies on the same seed face identical potential outcomes (common random numbers; paired tests are valid).
- Exact scoring matches a manual log-probability computation; hidden states differ by layer; LoRA training lowers the loss and raises the probability of the trained label (`tests/test_hf_tiny.py`).
- Every prompt is inspectable with `deadeye show-prompt` and every decision is logged with the raw model output, the parsed action, the legal set and the oracle action.

3.2 Statistical protocol (pre-registered)

- Unit of analysis: the episode (seed). Primary outcome: normalised return. Secondary: illegal-action rate, oracle agreement, task metrics (regret, success, win), latency and tokens per decision.
- Point estimates: mean and interquartile mean; uncertainty: 95% bootstrap CI over episodes (2000 resamples). Paired comparisons (same seeds): permutation test on sign flips of the paired differences, 5000 permutations, Holm-corrected within each family of comparisons. Effect sizes: paired Cohen's d.
- Sample size: for a paired comparison, $n = ((z_{0.975} + z_{0.8}) * sd_diff / effect)^2$, implemented as `deadeye.metrics.episodes_needed(effect, sd)`. With a smallest effect of interest of 0.10, 100 episodes suffice only when the SD of the paired differences is at most 0.36; the pilot measures that SD per task and the pre-registration records it. Tasks whose episode outcome is a single -1/0/+1 game have SDs near 1 in normalised units, which is why tic-tac-toe uses 300 episodes and blackjack packs 20 hands into each of its 100 episodes (the SD of an episode mean falls with the square root of the number of hands). Where even that leaves the detectable effect above 0.10, the paper reports the smallest effect the design could have detected instead of claiming a null.
- Scale curves: fit $score = a + b * \log_{10}(params)$ per (family, method, task) by least squares with bootstrap CIs on `b`; H1 is supported if $b > 0$ with CI excluding 0 for `prompt_generate` and the slopes differ by task class; H4 is supported if `b` under `prompt_score` is less than half of `b` under `prompt_generate`.
- Exclusions: a cell is excluded only if the model fails to load or a run crashes; no exclusion on outcome. Every excluded cell is listed in the paper's appendix.

3.3 Leakage and confound controls

- Training seeds (probe, LoRA, few-shot demonstrations) start at 100000 and never overlap evaluation seeds (0-999).
- Prompt wording is frozen after Phase 1 (`prereg-v1` tag). Prompt-format factors are run as explicit ablations, not tuned per model.
- Action labels are compared under both `name` and `letter` encodings so that tokenisation and surface-form effects are measured, not hidden.
- The oracle is deterministic; "oracle agreement" is therefore a lower bound when several actions are optimal (gridworld, tictactoe), which is why return, not agreement, is the primary outcome.
- Quantised models report the catalogue parameter count, not the packed weight count, and are labelled by precision in every table.
- API-served models (`openai` backend) are non-deterministic across servers; results from them are reported in a separate table and never mixed into the scale curves.

3.4 What would make us abandon a hypothesis

Written in `docs/preregistration.md`. In brief: a confidence interval on the relevant slope or difference that includes 0, or an effect in the opposite direction, in the confirmatory sweep.

4. Compute budget

Decisions per model per method (sweep defaults): bandit 5000, contextual bandit 3000, loan 2000, gridworld about 1000, tictactoe about 400, blackjack about 600; roughly 12000 decisions. Per model: `prompt_generate` x2 (plain, CoT), `prompt_score`, `probe` (plus 100 training episodes), `lora_sft` (plus training) - roughly 60000 forward passes of 300-600 prompt tokens.

| Model size | Backend | Seconds per decision (approx.) | Hours per model (sweep) |
| ---| ---| ---| ---|
| 135M-0.6B | CPU laptop | 0.2-0.8 | 4-13 |
| 0.5B-1.7B | one 24 GB GPU | 0.03-0.08 | 0.5-1.5 |
| 7B-14B | one 24 GB GPU (bf16) | 0.1-0.25 | 2-4 |
| 32B-72B | one 80 GB GPU (int4) | 0.4-1.0 | 7-17 |

The full sweep is about 150 GPU-hours on a single 80 GB GPU or 2-4 days on 2-4 GPUs. The five ablation blocks together are of the same order again, and they can be run and reported independently. `deadeye estimate` recomputes these numbers for any edit. See `docs/compute_budget.md`.

```

### ### 5. Threats to validity (and what the design does about them)

1. **Prompt sensitivity.** Mitigated by freezing prompts before the sweep and running format ablations; residual sensitivity is reported, not hidden.
2. **Tokenisation and surface-form competition in scoring.** Mitigated by name/letter encodings and length normalisation as explicit factors.
3. **Contamination.** Instances are generated on the fly from seeds; no instance text exists on the web. Rules (blackjack basic strategy, tic-tac-toe) are of course known to models; this is a feature of knowledge-driven decision-making, and the contextual bandit and sign-flipped loan variants are the tasks where prior knowledge cannot help.
4. **Cross-family confounds.** Scale claims are made within families only; cross-family plots are descriptive.
5. **Oracle quality.** Every oracle is optimal for the configured opponent or rules (expectimax against the random tic-tac-toe opponent, minimax against the minimax opponent, dynamic-programming-optimal hit/stand in blackjack, BFS in gridworld), but blackjack has no doubling or splitting, and oracle agreement is a lower bound wherever several actions are equally good. ``docs/known_limitations.md`` keeps the full list.
6. **Hardware non-determinism.** Greedy decoding and seeded sampling; the manifest records torch, transformers, CUDA, and GPU model for every run.

---

### ### 6. The end state ("done" checklist)

- [ ] ``prereg-v1`` tag with frozen prompts and ``docs/preregistration.md``
- [ ] ``results/sweep_gpu`` and ``results/ablations`` complete (no missing cells in ``run_manifest.json``)
- [ ] ``deadeye report ... --paper-dir paper`` regenerates every figure and table in the paper
- [ ] Paper: abstract, introduction, related work (from ``docs/literature_review.md``), method, results (H1-H7 each answered with a number and a CI), limitations, reproducibility statement, appendix with all prompts and the cell list
- [ ] ``v1.0`` tag, Docker image, Zenodo DOI for code + results, leaderboard HTML published
- [ ] Peer-testing instructions verified by one person who did not write the code (``docs/peer_testing.md``)

---

### ### 7. The two comparisons the paper makes

The paper's results have two parts, and the second was asked for explicitly: first, small against large open-weight models; second, small open-weight models (converted here) against the purpose-built decision models that appeared in September and October 2026.

#### #### 7.1 Small versus large open-weight models

Within each family ladder, every method and task: decision quality (normalised score, oracle agreement, task metrics), reliability (illegal-action rate), speed (latency per decision, median and 95th percentile, measured on one machine), efficiency (prompt and generated tokens per decision; an estimate of floating-point operations as 2 x parameters x tokens; peak memory at the recorded precision), and the quality-per-second and quality-per-parameter frontiers. The scale slopes (``slopes.csv``) and the cost frontier (``fig_pareto``) summarise it; H1, H2, H4, H5 and H7 are the pre-registered tests.

#### #### 7.2 Small open-weight conversions versus decision models

"Decision models" take a state and typed questions and return probabilities over allowed answers without generating text. The systems and what is verified about them (official cards, announcements and repositories; all performance numbers below are the makers' own):

System	Maker, licence	Base and size	How it decides	How we run it
---	---	---	---	---
Jev	TypeSafe AI; closed, hosted	undisclosed	"System One" model trained with reinforcement learning for calibrated decisions on synthetic data (maker's description)	hosted API, if access is granted; reference row only
Kev	0.8B / 4B / 9B / 27B	Jared Palmer; Apache-2.0	Qwen3.5-0.8B/4B/9B-Base (rank-16 LoRA + pointer head); Qwen3.8-27B (full fine-tune)	trained on a 10k-example decision set plus generated policy examples; System One-compatible server   local server ( <code>`kev.serve`</code> )
Clef	flash 9B / Clef 27B	Cloudflare; Apache-2.0	Qwen3.5-9B / Qwen3.8-27B, frozen	routing head + low-rank adapters; Jev-compatible   Workers AI, or the weights
pplx-decider	v1.1 27B	Perplexity; Apache-2.0 reported, gated repo	Qwen3.8-27B fine-tune	Decisions API (own schema)   hosted API through an adapter
Laya	Laya-multilingual / Laya-typed-decisions	Convai Innovations; Apache-2.0	ModernBERT-large 421M / mmBERT-base 322M encoders + decision head	Choice / Score / Noul primitives, one forward pass   local package
SemIf	(formerly OpenJev)	Theo Lee; MIT code	frozen Qwen3 0.6B, MiniCPMs 2B, Qwen3.5-4B (any checkpoint)	reads option logits in one forward pass, no training   local; and reproduced by our <code>`score_letter`</code> on the same checkpoint
Strands Decider	2B	Amazon	2B	open "Jev clone" (reported)   not run in v1

On our side: the same Qwen3.5 bases Kev and Clef are built on (0.8B, 4B, 9B), plus the small Qwen2.5 and Qwen3 checkpoints from the ladders, converted by scoring, probe and LoRA with oracle-labelled states.

Every system is reduced to the same numbers, measured on the same tasks and seeds: normalised score per task and its mean and worst task; oracle agreement; calibration of the probability it assigns to its chosen action against the oracle (expected calibration error and Brier score, which is where "calibrated decisions" is tested rather than assumed); latency per decision (median, 95th percentile, network included for hosted systems and reported separately); tokens and parameters; and openness (weights, code, training data, licence). The report

writes this as `system\_comparison.csv` and the paper prints it as one table with the calibration tables beside it.

Two of the rows double as validations. SemIf's readout is our letter-scoring condition, so SemIf on Qwen3.5-4B and our `score\_letter` on the same checkpoint should agree up to prompt wording. Kev on a Qwen3.5 base and our LoRA conversion of the same base differ only in training data (theirs: classification corpora; ours: oracle labels of the task), which isolates what the training target contributes.

What this comparison cannot do is control the closed system: Jev's data and architecture are unknown, so it is a reference row, not a tested hypothesis. It is reported as exploratory (RQ5) and kept out of the pre-registered tests.

### ### 8. Extensions after v1.0

- Preference / RL fine-tuning (DP0 on oracle-vs-model action pairs, GRP0 with environment reward) as a sixth conversion method; the `hf` backend already exposes everything needed.
- Text-game environments (TextWorld / ALFWorld / BabyAI-text) for long-horizon language tasks.
- Multi-agent and negotiation tasks; human-behaviour targets (Centaur-style) instead of oracles.
- Energy measurement per decision (NVML) to replace latency in the Pareto plots.

### # Pre-registration

Status: **\*\*draft\*\*** (to be frozen with the `prereg-v1` git tag after the Phase 1 pilot; any later change is recorded in the "Deviations" section with a date and reason).

### ### 1. Title

Do small open-weight language models make good decision-makers? A controlled study of scale, conversion method and task structure.

### ### 2. Research questions

RQ1. How does decision quality scale with parameter count within a model family, and does the slope depend on the task's decision structure?

RQ2. Does the conversion method (prompting, likelihood scoring, hidden-state probe, LoRA behaviour cloning) change the dependence on scale?

RQ3. Which other factors (instruction tuning, quantisation, reasoning budget, observation format, prompt encoding, adaptation data budget) move decision quality, and by how much?

RQ4. What is the quality / inference-cost frontier?

RQ5 (exploratory, added 8 October 2026, not a pre-registered test). How do small open-weight models converted here compare with purpose-built decision models (Kev, Clef, Laya, Perplexity's decider, SemIf, and the hosted Jev as a reference) in decision quality, calibration, latency and cost on the same tasks and seeds?

### ### 3. Hypotheses, predictions and falsification criteria

Each prediction names the analysis that tests it. "Score" means normalised return (random = 0, oracle = 1). CI means a 95% bootstrap confidence interval. Slopes are from  $\text{score} = a + b \log_{10}(\text{params})$  fitted within one family on the sweep results.

#	Prediction	Supported if	Falsified if
1	Under `prompt_generate`, $b > 0$	in every family	CI of $b$ excludes 0 and is positive for every family with $\geq 4$ sizes   any family's CI includes 0 or is negative
H1b	Slope is larger for gridworld and tictactoe than for loan	bootstrap CI of $(b_{\text{planning}} - b_{\text{loan}})$ excludes 0	CI includes 0
H2a	Instruct > base under `prompt_generate` at every size	paired difference CI > 0 at $\geq 80\%$ of sizes	fewer than 50%
H2b	The instruct - base gap under `probe` and `lora_sft` is < 0.05	CI of the mean gap lies within [-0.05, 0.05]	CI excludes that interval
H3a	Best `lora_sft` model $\leq 1.7B$ $\geq$ `prompt_generate` model 10x larger (in-distribution)	paired difference CI $\geq 0$ on at least 5 of 7 tasks	3 or fewer tasks
H3b	On `loan_sign_flip` and `loan_covariate_shift`, the small adapted model's loss relative to in-distribution is larger than the large model's	CI of the interaction excludes 0	CI includes 0
H4a	Illegal-action rate under `prompt_score` is 0	by construction	n/a
H4b	$b_{\text{score}} < 0.5 b_{\text{generate}}$	bootstrap CI of the ratio < 0.5	CI includes 0.5 or above
H5	int4 cost < 0.05 at $\geq 7B$ ; > 0.10 below 2B	CI of the paired bf16 - int4 difference meet both thresholds	either fails
H6a	No model under prompting reaches UCB1's bandit score	all CIs below the UCB1 score	any CI includes or exceeds it
H6b	Summary history > raw history for models < 2B	paired CI > 0 for all such models	any CI includes 0
H7a	Thinking / CoT improves gridworld + tictactoe at $\geq 4B$	paired CI > 0	CI includes 0
H7b	Thinking / CoT does not improve models < 2B	paired CI includes 0 or is negative	CI > 0
H7c	Latency-normalised score is lower with thinking at every size	score/latency ratio lower for every size	any size higher

### ### 4. Design

- Environments, parameters and seeds exactly as in `configs/sweep\_gpu.yaml`, `configs/sweep\_controls.yaml` and the five files in `configs/ablations/` at the `prereg-v1` tag.
- Evaluation seeds 0-99 (tic-tac-toe 0-299; blackjack 0-99 with 20 hands per episode); training seeds

100000-100099; the pilot used seeds 5000-5019 and is not part of the confirmatory data. Prompts frozen at the tag.

- Models: the families and sizes listed in the two configs; a model that cannot be loaded is dropped and listed in the appendix.
- Primary outcome: normalised return per episode. Secondary: illegal-action rate, oracle agreement, task metrics, latency per decision (mean, median, 95th percentile) and tokens per decision, and for every policy that assigns probabilities (scoring, probes, decision models) the expected calibration error and Brier score of the probability assigned to the chosen action against the oracle.

### ### 5. Sample size

For a paired comparison with smallest effect of interest 0.10,  $\alpha = 0.05$  (two-sided) and power 0.8, the required number of paired episodes is  $n = ((1.96 + 0.84) * \text{sd\_diff} / 0.10)^2$ , where  $\text{sd\_diff}$  is the standard deviation of the per-episode paired differences in normalised score. One hundred episodes therefore suffice when  $\text{sd\_diff} \leq 0.36$ . From the pilot (seeds 5000-5019),  $\text{sd\_diff}$  per task (to be filled in at the tag): bandit \_\_, contextual bandit \_\_, loan \_\_, gridworld \_\_, tictactoe \_\_, blackjack \_\_. Tic-tac-toe uses 300 episodes and blackjack 100 episodes of 20 hands because single-game outcomes are  $-1/0/+1$  and their normalised SD is near 1. For any task whose measured  $\text{sd\_diff}$  implies a detectable effect above 0.10 at the chosen  $n$ , we report the smallest detectable effect alongside the estimate and do not interpret a non-significant result as evidence of no effect.

### ### 6. Analysis plan

Per cell: mean, IQM, 95% bootstrap CI (2000 resamples). Paired comparisons: permutation test on sign-flipped paired differences (5000 permutations). Holm correction is applied within each hypothesis's family of tests, which is the set of `--pair` arguments of one `deadeye compare` call: H2a is one family per model family, H3a one family per task set, H5 one family per method, H6b and H7a-b one family each. Scale slopes (H1, H4b): ordinary least squares of cell-mean normalised score on  $\log_{10}(\text{parameters})$  within a family, method and task, with 95% CIs from 1000 bootstrap resamples of episodes within cells (`slopes.csv` from `deadeye report`; slope ratios from `slope_ratio`). All computed by `deadeye report` and `deadeye compare`; the exact code at the `prereg-v1` tag is the analysis.

### ### 7. Exclusion and stopping rules

No episode is excluded for its outcome. A cell is excluded only on load failure or crash (recorded in `run_manifest.json`). No interim analysis changes the design; the sweep runs to completion.

### ### 8. Deviations (append-only)

Date	What changed	Why
---	---	---

### # Testing as a peer

The goal of this page is that someone who did not write the code can (a) verify the harness, (b) reproduce a reported number, (c) evaluate their own model, and (d) submit results.

### ### A. Verify the harness (no downloads, 5 minutes)

```

```bash
git clone https://github.com/CaptiveON/DeadEye && cd DeadEye
python -m venv .venv && source .venv/bin/activate
pip install torch --index-url https://download.pytorch.org/whl/cpu # or your CUDA build
pip install -e ".[hf,dev]" tabulate
pytest -q # ~1-2 min: environments, parsing, runner, exact scoring, LoRA, report
deadeye smoke # ~30 s: mock + tiny random model through every env and method

```

`deadeye smoke` prints the path of an HTML report. Open it: the random row scores 0 and the oracle row scores 1 on every environment; the tiny random model is at chance.

B. Reproduce a reported number

Every number in the paper comes from a `summary.json` in a results directory with the config that produced it in `run_manifest.json`. To re-run one cell:

```

-force deadeye run configs/sweep_gpu.yaml --only-env loan --only-model Qwen/Qwen2.5-1.5B-Instruct --only-method prompt_score -
deadeye report results/sweep_gpu --out report/check
deadeye compare results/sweep_gpu --pair "loan:Qwen__Qwen2.5-1.5B-Instruct/prompt_score vs loan:Qwen__Qwen2.5-1.5B-Instruct/prompt_generate"

```

Greedy decoding and seeded instances make `prompt_score`, `probe` and `prompt_generate` (temperature 0) bit-reproducible on the same hardware and library versions, and statistically reproducible across hardware (compare CIs, not point values).

C. Evaluate your own model

1. **Local weights** (any `AutoModelForCausalLM`): add to a config

```
models:
- {backend: hf, id: your-org/your-model, dtype: bfloat16, params: 3000000000, family: yours, instruct: true}
```

params and family are optional for unquantised models (they are counted), required for quantised ones and for the scale plots.

2. A served model (vLLM, llama.cpp server, Ollama, LM Studio, OpenRouter):

```
models:
- {backend: openai, id: served-name, base_url: "http://localhost:8000/v1", params: 7000000000, family: yours}
methods:
- {name: prompt_generate, params: {max_new_tokens: 16}}
- {name: prompt_score, params: {action_format: letter}, label: score_letter}
```

Served models support prompt_generate and first-token prompt_score; probes and LoRA need local weights.

3. Run `deadeye run your.yaml` and `deadeye report results/yours`.

D. Add an environment or a method

- Environment: subclass `deadeye.envs.base.Environment`, implement `reset`, `step`, `oracle_action`, `describe`, set `action_labels` and `max_steps`, register it in `deadeye/envs/registry.py`, and add it to `tests/test_envs.py` (the parametrised tests cover legality, determinism and oracle > random automatically).
- Method: subclass `deadeye.policies.base.Policy` (see `policies/probe.py` for an offline-trained example) and register it in `deadeye/policies/registry.py`.

E. Submit results to the leaderboard

Open a pull request that adds `results/<your-run>/**/summary.json` and `episodes.jsonl` (not `steps.jsonl`, which can be large) plus the config. CI regenerates the leaderboard from all `results/` directories. State hardware, library versions (already in `run_manifest.json`) and whether prompts were modified.

F. What to look at when something seems wrong

- `steps.jsonl` has the raw model output for every decision; filter `"legal": false` to see format failures and `"parsed_action"` to see what the parser extracted.
- `deadeye show-prompt <env> --seed N --steps K` prints the exact messages.
- `prepare_stats` in `summary.json` reports probe training accuracy and LoRA loss before/after.

Compute budget

`deadeye estimate <config> --seconds-per-decision S` counts decisions from oracle rollouts and multiplies by models x methods. Use it after every config edit. Rules of thumb measured on the tiny model and typical hardware (prompts are 300-600 tokens, answers 1-16 tokens):

Setting	prompt_score (one forward, batched choices)	prompt_generate 16 tokens	embed (probe)
135M-360M, CPU (4 cores)	0.15-0.4 s	0.3-0.8 s	0.1-0.3 s
1.5B, CPU	1-2 s	2-4 s	0.8-1.5 s
1.5B, 24 GB GPU bf16	0.03 s	0.08 s	0.02 s
7B, 24 GB GPU bf16	0.08 s	0.25 s	0.06 s
14B, 24 GB GPU bf16	0.15 s	0.5 s	0.12 s
32B / 72B, 80 GB GPU int4	0.4 / 0.9 s	1.5 / 3 s	0.3 / 0.7 s

Chain-of-thought (`max_new_tokens: 256`) is roughly 10x `prompt_generate`; thinking mode with `max_new_tokens: 2048` can be 50x. Budget those cells explicitly.

LoRA: 100 training episodes yield 1000-5000 examples per environment; two epochs at batch 8 take 2-10 minutes on a GPU for 7B (QLoRA for int4 models), under a minute for sub-1B models.

Sweep (`configs/sweep_gpu.yaml`, 33 model entries, 6 methods): about 150 GPU-hours on one 80 GB GPU. The control ladders (`configs/sweep_controls.yaml`) add about 60. The five ablation blocks in `configs/ablations/` are 150-200 cells each; together they are of the same order as the sweep, and each can be run and reported on its own. Check any edit with `deadeye estimate <config>`. The CPU pilot (`configs/pilot_cpu.yaml`) is 6-12 hours on a laptop, or 25 minutes for the 135M model alone.

Memory: bf16 needs ~2 bytes/parameter plus KV cache (prompts are short); int4 ~0.6 bytes/parameter.

Known limitations

Found during the adversarial review of the harness; kept here so the paper's limitations section and the pre-registration stay honest. Items marked *decision* are deliberate and documented rather than bugs.

Environments

- **Tic-tac-toe oracle.** Against the random opponent the oracle is expectimax (the policy that maximises expected outcome against uniformly random replies); against the minimax opponent it is minimax. The minimax opponent breaks ties at random (seeded), so a deterministic policy meets many distinct games. Oracle agreement remains a lower bound wherever several moves are equally good.
- **Blackjack oracle.** Basic strategy for hit/stand with an infinite deck is the dynamic-programming optimum for these rules (expected value about -0.026 per hand), but the game has no doubling or splitting. Player and dealer cards come from separate pre-drawn streams, so pairing across policies is tight; the variance per hand is still large relative to the random-oracle span, which is why blackjack uses 300 episodes.
- **Contextual bandit and the feature-probe control.** The hidden weights are redrawn every seed and the feature vector is the current context only, so a classifier trained across seeds is at chance by construction. That control therefore measures nothing on this task; the task exists to test in-context learning, which the probe cannot do. *Decision.*
- **Few-shot demonstrations in tic-tac-toe** come from other seeds and may show the agent playing the other mark; each demonstration states its own mark in the observation text. *Decision.*
- **Clairvoyant bandit oracles.** On the bandit and contextual bandit the oracle knows the hidden means or weights, so it is an upper anchor rather than a policy any agent could implement from the text; UCB1 marks the achievable reference, and probes or LoRA trained on oracle labels learn a history-based approximation. *Decision.*
- **Blackjack episodes** can bundle several hands (`hands_per_episode`, 20 in the sweep) so that the per-episode normalised score has a workable variance; each hand is still a separate decision problem and the oracle is unchanged.

Methods and runner

- **Training caps.** Probes and LoRA stop collecting states at `max_train_states` (default 50000); `prepare_stats` records `n_train_episodes_used` and truncated so a silently smaller data budget is visible in the logs.
- **Out-of-distribution loan tasks.** `train_params` lets adaptation methods train on the in-distribution variant and evaluate on the shifted one; `configs/ablations/tasks.yaml` uses it. Without it, probes and LoRA trained on the shifted task are not out of distribution.
- **Terminating on illegal actions** (`illegal_action: terminate`) cuts episodes short and makes regret look better for early failures; no shipped config uses it. *Decision.*
- **Served models.** An OpenAI-compatible server applies its own sampling defaults, and first-token scoring resolves ties at the floor in favour of the first option. Results from served models are reported separately and never enter the scale curves. *Decision.*

Decision-model comparison

- **Jev is closed.** Its weights, architecture and training data are not public, so none of the controls used for the open ladders apply to it, and its latency includes a network round trip. The comparison in `configs/decision_models.yaml` is exploratory and stays out of the pre-registered hypotheses. *Decision.*
- **Protocol mapping.** The decision backend sends the task description and observation as the state and the legal actions as the criteria of one choice question, with a fixed instruction; a decision model never sees few-shot demonstrations or a chain-of-thought budget, so only the scoring condition applies to it.
- **Base-model identity.** The open decision models are built on Qwen3.5 and Qwen3.8 checkpoints whose repo ids must be confirmed on the hub before the run; the catalogue does not list them as verified.
- **Protocol adapters.** Kev and Clef speak the System One protocol the decision backend implements. Laya's package and Perplexity's Decisions API have their own schemas and need a thin adapter (same class, different request and response mapping) before they can be run; SemIf is reproduced exactly by `score_letter` on the same checkpoint, so its row can be filled either from its server or from ours.
- **Calibration scope.** ECE and Brier are computed on the probability of the action taken, against the oracle's action; where several actions are optimal the oracle's tie-break makes a well-calibrated policy look over-confident, so calibration is compared across systems within a task, never across tasks.

Statistics

- Normalisation anchors (random and oracle means) are treated as fixed; their sampling uncertainty is not propagated into the confidence intervals. With 100 episodes it is small relative to the per-cell intervals, and the paired tests do not depend on the anchors at all.
- Holm correction is applied within one report call, which treats every comparison in that report as a single family; the pre-registration names the families per hypothesis and the paper applies the correction per family with `deadeye compare`.

References audit

Audit date: 8 October 2026. Scope: the 106 entries of `paper/refs.bib` as committed in 57493f6, plus 7 methods-section references added at the coordinator's request. Target style: `natbib` (`authoryear`, `round`, `sort`) with `agsm.bst` (Harvard). Keys were never renamed.

Method

- **Primary sources.** Every entry was checked against a publisher or proceedings record: `proceedings.neurips.cc` and `neurips.cc`, `iclr.cc` / `proceedings.iclr.cc` and OpenReview, `proceedings.mlr.press` (ICML, AISTATS, UAI), `aclanthology.org`, `proceedings.mlsys.org`, `ojs.aaai.org`, `colmweb.org`, `link.springer.com`, `nature.com`, `PubMed/PMC` (PNAS, Nature), `tandfonline.com`, `jstor.org`, `jmlr.org`, `cambridge.org`, `mitpress.mit.edu`, `mheducation.com`, `WorldCat`; arXiv abstract pages for preprints and technical reports; and the releasing organisation's own pages for model documentation (GitHub organisations QwenLM, meta-llama, huggingface, EleutherAI, AI-Efficiency; `blog.google`; `developers.googleblog.com`; `mistral.ai`; `huggingface.co/blog`). Direct fetches from this container were blocked for most hosts, so confirmation came from search results restricted to those publisher domains and from direct reads of `GitHub/raw.githubusercontent.com`. The page used for each entry is in the table below.
- **Fields confirmed.** Author list (order and spelling), exact title, year, venue, volume/issue/pages where the venue prints them, DOI and arXiv identifier. Where a field could not be confirmed on a primary record it was removed, not guessed: the issue number of Auer et al. (2002), the page ranges of Henderson et al. (2018) and AWQ (MLSys 2024), and all NeurIPS page ranges (NeurIPS shows them only in a BibTeX download that was not reachable). NeurIPS DOIs were added only where the proceedings page itself showed one (Wei et al. 2022; Towers et al. 2025).

- **Published versions.** Every arXiv-only entry was searched for a peer-reviewed version (OpenReview, proceedings and journal sites). Upgraded entries keep the arXiv identifier in eprint; where the published title differs from the arXiv title, the arXiv title is kept in note.
- **Titles when venue metadata and paper disagree.** For three NeurIPS papers the proceedings index title differs from the title printed on the paper (GTBench: "Capabilities" vs "Limitations"; Hoffmann et al. 2022: "An Empirical Analysis of Compute-Optimal ..." vs "Training Compute-Optimal ..."; Dettmers et al. 2022: "GPT3.int8()" vs "LLM.int8()"). The bib uses the proceedings index title (as NeurIPS's own citation export and DBLP do) and gives the printed/arXiv title in note.
- **Author lists.** "and others" was replaced by the full list wherever the list has 30 names or fewer (12 entries). It is kept only where the full list exceeds 30 names: Kadavath et al. (36), Binz et al. 2025 (40), Qwen2.5 (42 named), Qwen3 (60), OLMo 2 (COLM version, Team OLMo + 42), Olmo 3 (67), Llama 3 (559) and the Nature DeepSeek-R1 paper (more than 100). Corporate authors are used where the document is issued by the organisation (Gemma Team, OpenAI, Google, Meta, Mistral AI, Qwen Team).
- **Format for agsm.bst.** agsm sentence-cases titles, so acronyms and proper nouns are braced; a sentence-case rendering of every title was checked programmatically (0 unprotected acronyms). agsm does not print doi, eprint or isbn (kept for the record) and prints url through natbib's \harvardurl, which typesets it with \textit rather than \url. A URL containing _, %, #, & or ~ would therefore break or corrupt the build, so url is used only for web documents (all @misc entries) and TMLR papers, all with URLs free of those characters. Web documents carry howpublished, url and note = {Accessed 8 October 2026}. AAAI papers are typed @article (journal *Proceedings of the AAAI Conference on Artificial Intelligence*, volume(issue)) because agsm warns when an @inproceedings has both volume and number.
- **Parser check.** bibtexparser 1.4.3 plus a strict custom checker: 113 entries, 113 unique keys, all 106 original keys present, braces balanced in every entry and field, all agsm-required fields present for each type, no unescaped & % # _ in printed fields, page ranges with --, no @ outside entries, url fields safe for \harvardurl. Result: no problems. (No TeX toolchain is installed, so bibtex itself was not run.)
- **Keys used by the paper.** At 22:48 UTC on 8 October the sections cite 73 distinct keys, all present in the bib; the other 40 entries are used by docs/literature_review.md or were added for the methods section.

(a) Summary

	Count
Entries audited (original bib)	106
Kept	106
of which upgraded to a peer-reviewed published version	8
of which corrected (authors, title, venue, year, pages or DOI changed)	38
of which normalised only (full proceedings names, publisher, braces, eprint, url/access date)	59
of which unchanged	1
Removed	0
Added at the coordinator's request (each verified)	7
Entries now in paper/refs.bib	113

Upgraded to the published version: alain2016probes; costarelli2024gamebench; deepseekai2025r1; liu2025quanthurts; olmo2025olmo2; tajwar2025curious; wang2024lookattext; zheng2025qwen3quant.

Source types (all 113 entries): peer-reviewed (conference): 66; arXiv preprint from established lab: 11; peer-reviewed (journal): 11; official technical report or model card (arXiv): 9; official model documentation (organisation web page / repository): 9; academic book: 4; peer-reviewed workshop paper (non-archival): 2; official software release (DOI): 1.

(b) Entry-by-entry audit

"Normalised" means the venue was rewritten to the full proceedings name (e.g. "The Twelfth International Conference on Learning Representations (ICLR 2024)", "Advances in Neural Information Processing Systems 37 (NeurIPS 2024)", "Proceedings of the 42nd International Conference on Machine Learning" with PMLR series/volume), a publisher was added, and title braces were checked; author lists, titles and years in those rows were confirmed unchanged.

Key	Type of source	Venue (as now cited)	Verified at	What changed
abdin2024phi4	official technical report or model card (arXiv)	arXiv preprint arXiv:2412.08905	https://arxiv.org/abs/2412.08905	Normalised. All 27 authors confirmed; eprint added.
abdin2025phi4reasoning	official technical report or model card (arXiv)	arXiv preprint arXiv:2504.21318	https://arxiv.org/abs/2504.21318	Corrected. "and others" replaced by the full 23-author list.
agarwal2021precipice	peer-reviewed (conference)	Advances in Neural Information Processing Systems 34 (NeurIPS 2021)	https://proceedings.neurips.cc/paper/2021/file/f514cec81cb148559cf475e7426eed5e-Paper.pdf ; https://neurips.cc/virtual/2021/poster/26712	Normalised. Volume moved into booktitle; publisher added.

Key	Type of source	Venue (as now cited)	Verified at	What changed
alain2016probes	peer-reviewed workshop paper (non-archival)	5th International Conference on Learning Representations (ICLR 2017), Workshop Track	https://openreview.net/forum?id=HJ4-rAVtl ; https://openreview.net/pdf?id=HJ4-rAVtl	Upgraded. arXiv preprint -> ICLR 2017 Workshop Track (OpenReview HJ4-rAVtl); year 2016 -> 2017; arXiv id kept.
allal2025smollm2	official technical report or model card (arXiv)	arXiv preprint arXiv:2502.02737	https://arxiv.org/abs/2502.02737	Corrected. "and others" replaced by the full 22-author list (COLM acceptance not found, kept as arXiv).
auer2002ucb	peer-reviewed (journal)	Machine Learning 47, 235–256	https://link.springer.com/article/10.1023/A:1013689704352	Corrected. Issue number "2-3" removed (not shown on the Springer record); publisher added.
baldwin1956optimum	peer-reviewed (journal)	Journal of the American Statistical Association 51(275), 429–439	https://www.tandfonline.com/doi/abs/10.1080/01621459.1956.10501334	Added. At the coordinator's request; verified (DOI).
belcak2025slm	arXiv preprint from established lab	arXiv preprint arXiv:2506.02153	https://arxiv.org/abs/2506.02153	Normalised. No published version (NVIDIA Research position paper); eprint added.
belinkov2022probing	peer-reviewed (journal)	Computational Linguistics 48(1), 207–219	https://aclanthology.org/2022.cl-1.7/ ; https://aclanthology.org/2022.cl-1.7.bib	Normalised. Publisher added.
biderman2023pythia	peer-reviewed (conference)	Proceedings of the 40th International Conference on Machine Learning, PMLR 202, pp. 2397–2430	https://proceedings.mlr.press/v202/biderman23a.html	Corrected. Author name forms as in PMLR (Quentin Gregory Anthony); venue normalised.
biderman2024lessons	arXiv preprint from established lab	arXiv preprint arXiv:2405.14782	https://arxiv.org/abs/2405.14782 ; https://arxiv.org/html/2405.14782v2	Corrected. "and others" replaced by the full 30-author v2 list.
binz2023cognitive	peer-reviewed (journal)	Proceedings of the National Academy of Sciences 120(6), e2218523120	https://pubmed.ncbi.nlm.nih.gov/36730192/ ; https://www.pnas.org/doi/10.1073/pnas.2218523120	Unchanged. All fields confirmed (issue 6 on PMC); no change.
binz2025centaur	peer-reviewed (journal)	Nature 644(8078), 1002–1009	https://www.nature.com/articles/s41586-025-09215-4 ; https://pmc.ncbi.nlm.nih.gov/articles/PMC12390832/	Normalised. 40 authors (>30), so leading authors + "and others" kept; note reworded.
brown2020gpt3	peer-reviewed (conference)	Advances in Neural Information Processing Systems 33 (NeurIPS 2020)	https://proceedings.neurips.cc/paper/2020/file/1457c0d6bfcb4967418bf8ac142f64a-Paper.pdf ; https://neurips.cc/virtual/2020/public/poster_1457c0d6bfcb4967418bf8ac142f64a.html	Normalised. 31 authors confirmed; volume moved into booktitle; publisher added.
carta2023glam	peer-reviewed (conference)	Proceedings of the 40th International Conference on Machine Learning, PMLR 202, pp. 3676–3713	https://proceedings.mlr.press/v202/carta23a.html	Normalised. Venue normalised.
chen2021decision	peer-reviewed (conference)	Advances in Neural Information Processing Systems 34 (NeurIPS 2021)	https://proceedings.neurips.cc/paper/2021/hash/7f489f642a0ddb10272b5c31057f0663-Abstract.html	Corrected. Author name as in the NeurIPS record (Misha Laskin); title braces; volume moved into booktitle.

Key	Type of source	Venue (as now cited)	Verified at	What changed
chu2025sft	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 10818–10838	https://proceedings.mlr.press/v267/chu25c.html	Corrected. "and others" replaced by the full 9-author list; pages 10818–10838 added.
costarelli2024gamebench	peer-reviewed workshop paper (non-archival)	Language Gamification Workshop at the 38th Conference on Neural Information Processing Systems (NeurIPS 2024)	https://neurips.cc/virtual/2024/106557 ; https://neurips.cc/media/neurips-2024/Slides/106557.pdf ; https://arxiv.org/abs/2406.06613	Upgraded. arXiv preprint -> NeurIPS 2024 Language Gamification workshop paper (peer-reviewed, non-archival); arXiv id kept.
cuadron2025overthinking	arXiv preprint from established lab	arXiv preprint arXiv:2502.08235	https://arxiv.org/abs/2502.08235 ; https://sky.cs.berkeley.edu/project/agentthink/	Corrected. "and others" replaced by the full 16-author list.
deepseekai2025r1	peer-reviewed (journal)	Nature 645(8081), 633–638	https://www.nature.com/articles/s41586-025-09422-z ; https://pubmed.ncbi.nlm.nih.gov/40962978/	Upgraded. arXiv preprint -> Nature 645(8081):633–638 (2025). DOI added; author now the Nature byline (Guo et al., >100 names); arXiv id and title kept in note.
dettmers2022llmint8	peer-reviewed (conference)	Advances in Neural Information Processing Systems 35 (NeurIPS 2022)	https://proceedings.neurips.cc/paper_files/paper/2022/file/c3ba4962c05c49636d4c6206a97e9c8a-Paper-Conference.pdf ; https://neurips.cc/virtual/2022/poster/53018	Added. At the coordinator's request; verified. Proceedings index title "GPT3.int80..." used, printed title "LLM.int80..." in note.
dettmers2023kbit	peer-reviewed (conference)	Proceedings of the 40th International Conference on Machine Learning, PMLR 202, pp. 7750–7774	https://proceedings.mlr.press/v202/dettmers23a.html	Normalised. Venue normalised.
dettmers2023qlora	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	https://proceedings.neurips.cc/paper_files/paper/2023/hash/1feb87871436031bdc0f2beaa62a049b-Abstract-Conference.html	Normalised. Volume moved into booktitle; publisher added.
dilkes2025reinforced	arXiv preprint from established lab	arXiv preprint arXiv:2508.10839	https://arxiv.org/abs/2508.10839 ; https://eprints.soton.ac.uk/505168/	Normalised. No published version found; eprint added.
duan2024gtbench	peer-reviewed (conference)	Advances in Neural Information Processing Systems 37 (NeurIPS 2024)	https://proceedings.neurips.cc/paper_files/paper/2024/hash/3191170938b6102e5c203b036b7c16dd-Abstract-Conference.html	Normalised. Volume moved into booktitle (NeurIPS 37); publisher added; note reworded (index title vs PDF/arXiv title).
efron1993bootstrap	academic book	Chapman & Hall (Monographs on Statistics and Applied Probability 57)	https://search.worldcat.org/title/introduction-to-the-bootstrap/oclc/797437299 ; https://openlibrary.org/books/OL1397023M/An_introduction_to_the_bootstrap ; https://www.taylorfrancis.com/books/mono/10.1201/9780429246593/introduction-bootstrap-bradley-efron-tibshirani	Added. At the coordinator's request; verified (original Chapman & Hall imprint 1993; T&F DOI of the publisher's listing).
fan2024rational	peer-reviewed (conference)	Proceedings of the AAAI Conference on Artificial Intelligence 38(16), 17960–17967	https://ojs.aaai.org/index.php/AAAI/article/view/29751	Corrected. Pages 17960–17967 added; retyped as AAAI journal-style record (38(16)); title braces.

Key	Type of source	Venue (as now cited)	Verified at	What changed
frantar2023gptq	peer-reviewed (conference)	The Eleventh International Conference on Learning Representations (ICLR 2023)	https://iclr.cc/virtual/2023/poster/10855 ; https://openreview.net/forum?id=tcbBPnfwxS	Normalised. Venue normalised; note reworded.
gao2023lmeval	official software release (DOI)	Zenodo software release of EleutherAI's lm-evaluation-harness, version v0.4.0, 4 December 2023	https://zenodo.org/records/10256836 ; https://raw.githubusercontent.com/EleutherAI/lm-evaluation-harness/main/CITATION.bib	Normalised. Restructured as web/software record; howpublished, url and access date; non-standard version/publisher fields folded into howpublished.
gema2025inverse	peer-reviewed (journal)	Transactions on Machine Learning Research	https://openreview.net/pdf/16c9d2ec1a2c68ff255078dd243912d6f9fd1591.pdf ; https://openreview.net/forum?id=NXgyHW1c7M ; https://arxiv.org/abs/2507.14417	Corrected. "and others" replaced by the full 14-author list; TMLR OpenReview URL and ISSN added.
gemma2025gemma3	official technical report or model card (arXiv)	arXiv preprint arXiv:2503.19786	https://arxiv.org/abs/2503.19786	Normalised. Gemma Team corporate author kept (214 contributors); note and eprint added.
google2025gemma3n	official model documentation (organisation web page / repository)	Google Developers Blog, 26 June 2025	https://developers.googleblog.com/en/introducing-gemma-3n-developer-guide/ ; https://ai.google.dev/gemma/docs/releases	Normalised. Publication date (26 June 2025) added; url and access date restructured.
google2026gemma4	official model documentation (organisation web page / repository)	Google blog, 2 April 2026 (announcement by C. Farabet and O. Lacombe, Google DeepMind)	https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/ ; https://deepmind.google/blog/gemma-4-byte-for-byte-the-most-capable-open-models/	Normalised. Publication date (2 April 2026) and byline added; url and access date restructured.
grattafiori2024llama3	official technical report or model card (arXiv)	arXiv preprint arXiv:2407.21783	https://arxiv.org/abs/2407.21783	Normalised. 559 authors (>30), leading authors + "and others" kept; note shortened; eprint added.
guertler2025textarena	arXiv preprint from established lab	arXiv preprint arXiv:2504.11442	https://arxiv.org/abs/2504.11442	Normalised. No published version found; arXiv id added as eprint.
guo2017calibration	peer-reviewed (conference)	Proceedings of the 34th International Conference on Machine Learning, PMLR 70, pp. 1321–1330	https://proceedings.mlr.press/v70/guo17a.html	Normalised. Venue normalised.
harris2026explore	peer-reviewed (conference)	Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence, PMLR 337, pp. 2008–2058	https://proceedings.mlr.press/v337/harris26a.html	Corrected. Added pages 2008–2058 (PMLR 337).
hegselmann2023tabllm	peer-reviewed (conference)	Proceedings of the 26th International Conference on Artificial Intelligence and Statistics, PMLR 206, pp. 5549–5581	https://proceedings.mlr.press/v206/hegselmann23a.html	Normalised. Venue normalised; pages confirmed.
henderson2018matters	peer-reviewed (conference)	Proceedings of the AAAI Conference	https://ojs.aaai.org/index.php/AAAI/article/view/11694	Corrected. Retyped as AAAI journal-style

Key	Type of source	Venue (as now cited)	Verified at	What changed
		on Artificial Intelligence 32(1)		record (volume 32, issue 1); page range removed (not shown on the AAAI record).
hendrycks2021mmlu	peer-reviewed (conference)	The Ninth International Conference on Learning Representations (ICLR 2021)	https://openreview.net/forum?id=d7KBjmI3GmQ	Normalised. Venue normalised.
hf2025smollm3	official model documentation (organisation web page / repository)	Hugging Face blog (official release write-up of SmoLLM3)	https://huggingface.co/blog/smolm3 ; https://github.com/huggingface/blog/blob/main/smolm3.md ; https://huggingface.co/HuggingFaceTB/SmolLM3-3B	Corrected. Corporate author replaced by the 23 named authors of Hugging Face's own recommended citation; url and access date.
hoffmann2022chinchilla	peer-reviewed (conference)	Advances in Neural Information Processing Systems 35 (NeurIPS 2022)	https://proceedings.neurips.cc/paper_files/paper/2022/hash/c1e2faff6f588870935f114ebe04a3e5-Abstract-Conference.html ; https://neurips.cc/virtual/2022/poster/53031	Corrected. Author order and given names set to the NeurIPS proceedings index (Vinyals before Rae; Thomas Hennigan, Katherine Millican, Jack Rae); volume moved into booktitle; note reworded.
holm1979simple	peer-reviewed (journal)	Scandinavian Journal of Statistics 6(2), 65–70	https://www.jstor.org/stable/i412579 ; https://www.jstor.org/stable/4615733	Added. At the coordinator's request; verified.
holtzman2021surface	peer-reviewed (conference)	Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, pp. 7038–7051	https://aclanthology.org/2021.emnlp-main.564/	Corrected. DOI, publisher and address added.
hu2022lora	peer-reviewed (conference)	The Tenth International Conference on Learning Representations (ICLR 2022)	https://openreview.net/forum?id=nZeVKeeFYf9 ; https://iclr.cc/virtual/2022/poster/6319	Normalised. Venue normalised.
huang2022zeroshot	peer-reviewed (conference)	Proceedings of the 39th International Conference on Machine Learning, PMLR 162, pp. 9118–9147	https://proceedings.mlr.press/v162/huang22a.html	Corrected. Added PMLR series, volume 162 and pages 9118-9147.
jia2024decision	peer-reviewed (conference)	Advances in Neural Information Processing Systems 37 (NeurIPS 2024)	https://proceedings.neurips.cc/paper_files/paper/2024/hash/cda04d7ea67ea1376bf8c6962d8541e0-Abstract-Conference.html ; https://neurips.cc/virtual/2024/poster/93425	Normalised. Volume moved into booktitle; publisher added.
jiwatode2026spatial	arXiv preprint from established lab	arXiv preprint arXiv:2607.22732	https://arxiv.org/abs/2607.22732	Corrected. Unverified "slated for CoG 2026" replaced by the authors' own arXiv comment; no proceedings record yet.
kadavath2022know	arXiv preprint from established lab	arXiv preprint arXiv:2207.05221	https://arxiv.org/abs/2207.05221	Normalised. 36 authors (>30), so leading authors + "and others" kept; eprint added.

Key	Type of source	Venue (as now cited)	Verified at	What changed
kaplan2020scaling	arXiv preprint from established lab	arXiv preprint arXiv:2001.08361	https://arxiv.org/abs/2001.08361	Normalised. No published version (OpenAI); eprint added.
krishnamurthy2024explore	peer-reviewed (conference)	Advances in Neural Information Processing Systems 37 (NeurIPS 2024)	https://proceedings.neurips.cc/paper_files/paper/2024/hash/d951f73c521d069fefbb73396df01424-Abstract-Conference.html	Normalised. Volume moved into booktitle; publisher added.
kumar2025precision	peer-reviewed (conference)	The Thirteenth International Conference on Learning Representations (ICLR 2025)	https://iclr.cc/virtual/2025/oral/31737 ; https://iclr.cc/virtual/2025/poster/27832	Normalised. Venue normalised.
laskin2023incontext	peer-reviewed (conference)	The Eleventh International Conference on Learning Representations (ICLR 2023)	https://openreview.net/pdf?id=hy0a5MMPUv ; https://iclr.cc/virtual/2023/poster/11148	Normalised. Venue normalised.
lattimore2020bandit	academic book	Cambridge University Press	https://www.cambridge.org/9781108486828 ; https://www.cambridge.org/core/books/bandit-algorithms/8E39FD004E6CE036680F90DD0C6F09FC	Normalised. Address added.
law2015simulation	academic book	McGraw-Hill Education, Fifth edn	https://www.mheducation.com/highered/product/simulation-modeling-and-analysis-law.html	Added. At the coordinator's request (preferred common-random-numbers reference); verified; Glasserman and Yao (1992) not added.
li2025smallstruggle	peer-reviewed (conference)	Findings of the Association for Computational Linguistics: ACL 2025, pp. 25366–25394	https://aclanthology.org/2025.findings-acl.1301/	Corrected. "and others" replaced by the full 8-author list; pages 25366–25394, DOI, publisher and address added.
lin2024awq	peer-reviewed (conference)	Proceedings of Machine Learning and Systems 6 (MLSys 2024)	https://proceedings.mlsys.org/paper_files/paper/2024/hash/42a452cbafa9dd64e9ba4aa95cc1ef21-Abstract-Conference.html	Normalised. Venue normalised (MLSys 2024, vol. 6); page range not added (seen only in a citing paper).
lin2024urial	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	https://proceedings.iclr.cc/paper_files/paper/2024/hash/6bcb4a501dbad0eba1b660c1a55318c-Abstract-Conference.html	Normalised. Venue normalised.
liu2024agentbench	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	https://proceedings.iclr.cc/paper_files/paper/2024/file/e9df36b21ff4ee211a8b71ee8b7e9f57-Paper-Conference.pdf ; https://iclr.cc/virtual/2024/poster/17388	Normalised. All 22 authors confirmed; venue normalised.
liu2025mindstep	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 38489–38517	https://proceedings.mlr.press/v267/liu25t.html	Normalised. Venue normalised.
liu2025quanthurts	peer-reviewed (conference)	Second Conference on Language Modeling (COLM 2025)	https://colmweb.org/2025/AcceptedPapers.html ; https://openreview.net/forum?id=BM192Ps5Nv	Upgraded. arXiv preprint -> COLM 2025 (OpenReview BM192Ps5Nv); arXiv id kept.
liu2025rational	peer-reviewed (conference)	The Thirteenth International Conference on Learning	https://proceedings.iclr.cc/paper_files/paper/2025/hash/7cdb411d07bdf6cd43ddf1254fae98cd-Abstract-Conference.html ; https://openreview.net/forum?id=dAeET8gxgq	Normalised. Venue normalised.

Key	Type of source	Venue (as now cited)	Verified at	What changed
		Representations (ICLR 2025)		
liu2026spiral	peer-reviewed (conference)	The Fourteenth International Conference on Learning Representations (ICLR 2026)	https://iclr.cc/virtual/2026/poster/10011289 ; https://proceedings.iclr.cc/paper_files/paper/2026/file/3ba7c8c36b71b2fc31fae3361aefefbf-Paper-Conference.pdf	Corrected. Author order corrected to the ICLR 2026 camera-ready (Yu second, Guertler fourth).
meta2025llamamodels	official model documentation (organisation web page / repository)	Official GitHub repository meta-llama/llama-models (model cards, release table and licences for Llama 3.1, 3.2, 3.3 and 4)	https://github.com/meta-llama/llama-models	Corrected. Invented descriptive title replaced by the repository title ("Llama Models"); url and access date restructured.
mistral2025mistral3	official model documentation (organisation web page / repository)	Mistral AI news announcement, 2 December 2025 (Mistral Large 3 and the Ministral 3 family)	https://mistral.ai/news/mistral-3/	Corrected. Title corrected to the page title "Introducing Mistral 3"; url and access date restructured.
mistral2025small3	official model documentation (organisation web page / repository)	Mistral AI news announcement, 30 January 2025	https://mistral.ai/news/mistral-small-3/	Normalised. Title braces; date, url and access date restructured.
mistral2025small31	official model documentation (organisation web page / repository)	Mistral AI news announcement, 17 March 2025	https://mistral.ai/news/mistral-small-3-1/	Normalised. Title braces; date, url and access date restructured.
monea2025icrl	peer-reviewed (conference)	Second Conference on Language Modeling (COLM 2025)	https://openreview.net/forum?id=c0RsezY2D1	Normalised. COLM 2025 confirmed on OpenReview; venue normalised.
nie2025evolve	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 46346–46376	https://proceedings.mlr.press/v267/nie25b.html	Corrected. Author name form as in PMLR (Jonathan Lee); venue normalised.
olmo2025olmo2	peer-reviewed (conference)	Second Conference on Language Modeling (COLM 2025)	https://openreview.net/pdf?id=2ezugTT9kU ; https://colmweb.org/2025/AcceptedPapers.html ; https://arxiv.org/abs/2501.00656	Upgraded. arXiv preprint -> COLM 2025 "2 OLMo 2 Furious (COLM's Version)" (OpenReview 2ezugTT9kU); author list as in COLM (Walsh et al., >30); arXiv id kept.
olmo2025olmo3	official technical report or model card (arXiv)	arXiv preprint arXiv:2512.13961	https://arxiv.org/abs/2512.13961	Corrected. Author list truncated to the confirmed leading entries (Team Olmo, Ettinger; 67 names, >30); eprint added.
openai2025gptoss	official technical report or model card (arXiv)	arXiv preprint arXiv:2508.10925	https://arxiv.org/abs/2508.10925	Normalised. OpenAI corporate author kept (124 named contributors); eprint added.

Key	Type of source	Venue (as now cited)	Verified at	What changed
orgad2025know	peer-reviewed (conference)	The Thirteenth International Conference on Learning Representations (ICLR 2025)	https://proceedings.iclr.cc/paper_files/paper/2025/hash/a712d461e57201efe35d429a6f1731c1-Abstract-Conference.html	Corrected. "and others" replaced by the full 7-author list.
paglieri2025balrog	peer-reviewed (conference)	The Thirteenth International Conference on Learning Representations (ICLR 2025)	https://iclr.cc/virtual/2025/poster/28856 ; https://proceedings.iclr.cc/paper_files/paper/2025/file/f0b1515be276f6ba82b4f2b25e50bef0-Paper-Conference.pdf	Corrected. Author spelling Wolczyk -> Wolczyk; Foerster given name as in the ICLR record; venue normalised.
pedregosa2011scikit	peer-reviewed (journal)	Journal of Machine Learning Research 12(85), 2825–2830	https://jmlr.org/papers/v12/pedregosa11a.html	Added. At the coordinator's request; verified (16 authors).
pomerleau1988alvinn	peer-reviewed (conference)	Advances in Neural Information Processing Systems 1 (NIPS 1988), pp. 305–313	https://proceedings.neurips.cc/paper/1988/file/812b4ba287f5ee0bc9d43bbf5bbe87fb-Paper.pdf ; https://proceedings.neurips.cc/paper/1988 ; https://www.ri.cmu.edu/publications/alvinn-an-autonomous-land-vehicle-in-a-neural-network/	Corrected. Editor (D. S. Touretzky) and publisher (Morgan Kaufmann) added; pages confirmed; note on the 1989 volume date.
qwen2024qwen25	official technical report or model card (arXiv)	arXiv preprint arXiv:2412.15115	https://arxiv.org/abs/2412.15115 ; https://github.com/QwenLM/Qwen3	Corrected. Author form follows the Qwen team's own citation (An Yang et al.; 42 named, >30); group name moved to note; eprint added.
qwen2025qwen3github	official model documentation (organisation web page / repository)	Official GitHub repository QwenLM/Qwen3 (model release notes, including the Qwen3-2507 updates)	https://github.com/QwenLM/Qwen3	Corrected. Invented descriptive title replaced by the repository title ("Qwen3"); url and access date restructured.
qwen2026github	official model documentation (organisation web page / repository)	Official GitHub repository QwenLM/Qwen3.8 (release notes for Qwen3.5, Qwen3.6 and Qwen3.8)	https://github.com/QwenLM/Qwen3.8	Corrected. Invented descriptive title replaced by the repository title ("Qwen3.8"); url and access date restructured.
rafaelov2023dpo	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	https://papers.nips.cc/paper/2023/hash/a85b405ed65c6477a4fe8302b5e06ce7-Abstract-Conference.html ; https://neurips.cc/virtual/2023/poster/72164	Normalised. Order Manning before Ermon confirmed on the NeurIPS abstract page; note removed.
robinson2023mcsb	peer-reviewed (conference)	The Eleventh International Conference on Learning Representations (ICLR 2023)	https://iclr.cc/virtual/2023/poster/10737 ; https://openreview.net/forum?id=yKbprarjc5B	Corrected. Author list set to the ICLR 2023 record (Robinson and Wingate); arXiv lists Rytting as well.
ross2011dagger	peer-reviewed (conference)	Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics, PMLR 15, pp. 627–635	https://proceedings.mlr.press/v15/ross11a.html	Corrected. PMLR series/volume 15 and address added; author name forms as in PMLR.
ruoss2025lmact	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning,	https://proceedings.mlr.press/v267/ruoss25a.html	Normalised. Confirmed PMLR 267:52303-52344; venue

Key	Type of source	Venue (as now cited)	Verified at	What changed
		PMLR 267, pp. 52303–52344		normalised, publisher added.
schaeffer2023mirage	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	https://proceedings.neurips.cc/paper_files/paper/2023/hash/ad98a266f45005c403b8311ca7e8bd7-Abstract-Conference.html	Normalised. Volume moved into booktitle; publisher added.
schmied2026greedy	peer-reviewed (conference)	The Fourteenth International Conference on Learning Representations (ICLR 2026)	https://proceedings.iclr.cc/paper_files/paper/2026/hash/cf2fcd40de8520027af2eadcafdec6-Abstract-Conference.html	Normalised. Venue normalised (ICLR 2026).
sclar2024formatting	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	https://proceedings.iclr.cc/paper_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html	Normalised. Venue normalised.
shao2024deepseekmath	arXiv preprint from established lab	arXiv preprint arXiv:2402.03300	https://arxiv.org/abs/2402.03300	Normalised. No published version (DeepSeek-AI); GRPO note removed; eprint added.
shinn2023reflexion	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	https://proceedings.neurips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html	Corrected. Author list set to the NeurIPS 2023 record (5 authors); Edward Berman appears only on arXiv v4. Note removed.
shojaee2025illusion	peer-reviewed (conference)	Advances in Neural Information Processing Systems 38 (NeurIPS 2025)	https://proceedings.neurips.cc/paper_files/paper/2025/hash/9b26ad15462c81548c0689188d2e8018-Abstract-Conference.html ; https://openreview.net/forum?id=YghiOusmvw	Normalised. Volume (NeurIPS 38) added to booktitle; publisher added.
sinha2026illusion	peer-reviewed (conference)	The Fourteenth International Conference on Learning Representations (ICLR 2026)	https://iclr.cc/virtual/2026/poster/10011614	Normalised. Venue normalised.
skean2025layer	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 55854–55875	https://proceedings.mlr.press/v267/skean25a.html	Corrected. "and others" replaced by the full 7-author list.
snell2025ttc	peer-reviewed (conference)	The Thirteenth International Conference on Learning Representations (ICLR 2025)	https://proceedings.iclr.cc/paper_files/paper/2025/file/1b623663fd9b874366f3ce019fd44-Paper-Conference.pdf ; https://iclr.cc/virtual/2025/oral/31924	Corrected. Title changed to the ICLR 2025 title ("... than Scaling Parameters for Reasoning"); arXiv title moved to note.
sutton2018rl	academic book	MIT Press, Second edn	https://mitpress.mit.edu/9780262039246/reinforcement-learning/	Normalised. Address added.
szot2024llarp	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	https://proceedings.iclr.cc/paper_files/paper/2024/file/8f9461d11e8ce9b0b28b21bfc645d74e-Paper-Conference.pdf ; https://openreview.net/pdf?id=u6imHU4Ebu	Normalised. Venue normalised.
tajwar2025curious	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 58227–58281	https://proceedings.mlr.press/v267/tajwar25a.html ; https://icml.cc/virtual/2025/poster/45106	Upgraded. arXiv preprint -> ICML 2025, PMLR 267:58227–58281; arXiv id kept.
tam2024speakfreely	peer-reviewed (conference)	Proceedings of the 2024 Conference on Empirical Methods in Natural	https://aclanthology.org/2024.emnlp-industry.91/	Normalised. Address and publisher added;

Key	Type of source	Venue (as now cited)	Verified at	What changed
		Language Processing; Industry Track, pp. 1218–1236		arXiv title recorded in note.
tan2024twosome	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	https://openreview.net/forum?id=hILVmJ4Uvu	Normalised. Venue normalised.
towers2025gymnasium	peer-reviewed (conference)	Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Datasets and Benchmarks Track	https://proceedings.neurips.cc/paper_files/paper/2025/hash/d7ff1795e8527f6443371c3933bdb52b-Abstract-Datasets_and_Benchmarks_Track.html ; https://neurips.cc/virtual/2025/poster/121446 ; https://github.com/Farama-Foundation/Gymnasium	Normalised. NeurIPS 38 D&B and DOI confirmed; title braces; publisher added.
wang2023selfconsistency	peer-reviewed (conference)	The Eleventh International Conference on Learning Representations (ICLR 2023)	https://openreview.net/forum?id=1PL1NIMMrw ; https://iclr.cc/virtual/2023/poster/11718	Normalised. Quoc V. Le initial; venue normalised.
wang2024lookattext	peer-reviewed (conference)	First Conference on Language Modeling (COLM 2024)	https://openreview.net/forum?id=qHdSA85GyZ ; https://colmweb.org/2024/AcceptedPapers.html	Upgraded. arXiv preprint -> COLM 2024 (OpenReview qHdSA85GyZ); arXiv id kept.
wang2024myanswerc	peer-reviewed (conference)	Findings of the Association for Computational Linguistics: ACL 2024, pp. 7407–7416	https://aclanthology.org/2024.findings-acl.441/	Corrected. "and others" replaced by the full 8-author list; DOI, publisher and address added.
wang2025ragen	arXiv preprint from established lab	arXiv preprint arXiv:2504.20073	https://arxiv.org/abs/2504.20073	Corrected. "and others" replaced by the full 18-author list.
wei2022cot	peer-reviewed (conference)	Advances in Neural Information Processing Systems 35 (NeurIPS 2022)	https://proceedings.neurips.cc/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html	Corrected. DOI 10.52202/068431-1800 added; Ed H. Chi as in the paper.
wei2022emergent	peer-reviewed (journal)	Transactions on Machine Learning Research	https://openreview.net/forum?id=vzkSU5zdWd ; https://arxiv.org/pdf/2206.07682	Normalised. TMLR OpenReview URL and ISSN added; journal name normalised.
wolf2020transformers	peer-reviewed (conference)	Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pp. 38–45	https://aclanthology.org/2020.emnlp-demos.6/	Added. At the coordinator's request; verified (22 authors, DOI).
wu2024smartplay	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	https://proceedings.iclr.cc/paper_files/paper/2024/hash/063d1fadbd557466bf5bcffe814a64e-Abstract-Conference.html	Normalised. Venue normalised to full ICLR 2024 name; publisher added.
yang2025qwen3	official technical report or model card (arXiv)	arXiv preprint arXiv:2505.09388	https://arxiv.org/abs/2505.09388 ; https://github.com/QwenLM/Qwen3	Normalised. 60 authors (>30), leading authors + "and others" kept; note and eprint added.
yao2023react	peer-reviewed (conference)	The Eleventh International Conference on Learning	https://openreview.net/forum?id=WE_vluYUL-X ; https://iclr.cc/virtual/2023/oral/12647	Normalised. Venue normalised; publisher added.

Key	Type of source	Venue (as now cited)	Verified at	What changed
		Representations (ICLR 2023)		
zhao2021calibrate	peer-reviewed (conference)	Proceedings of the 38th International Conference on Machine Learning, PMLR 139, pp. 12697–12706	https://proceedings.mlr.press/v139/zhao21c/zhao21c.pdf	Normalised. Venue normalised; internal note (PMLR lists "Zihao Zhao") moved to this audit.
zheng2024selectors	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	https://openreview.net/pdf?id=shr9PXz7T0	Normalised. Venue normalised.
zheng2025qwen3quant	peer-reviewed (journal)	Visual Intelligence 4, Article 11	https://link.springer.com/article/10.1007/s44267-026-00114-4 ; https://github.com/AI-Efficiency/Qwen3-Quantization-Toolkit	Upgraded. arXiv preprint -> Visual Intelligence 4, article 11 (Springer, 2026), DOI added; journal author list adds Zining Wang; year 2025 -> 2026; arXiv id kept.
zhou2023ifeval	arXiv preprint from established lab	arXiv preprint arXiv:2311.07911	https://arxiv.org/abs/2311.07911	Normalised. No published version (Google); eprint added.
zhou2023lima	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	https://proceedings.neurips.cc/paper_files/paper/2023/hash/ac662d74829e4407ce1d126477f4a03a-Abstract-Conference.html	Corrected. "and others" replaced by the full 15-author list.

(c) Removed

None. Every one of the 106 original entries was verified against a primary source, and every entry meets the source rule. 72 are peer-reviewed conference or journal papers (64 conference, 8 journal). 2 are peer-reviewed workshop papers. 2 are academic books. 11 are arXiv preprints from established groups (OpenAI, Anthropic, Google, NVIDIA Research, DeepSeek-AI, EleutherAI, UC Berkeley, Northwestern/Stanford/Microsoft, A*STAR/NUS/MIT-IBM, University of Southampton, Leibniz University Hannover) with no published version. 9 are technical reports or model cards on arXiv from the releasing organisation. 9 are official documentation pages or repositories of the releasing organisation, and 1 is a DOI-registered software release by its maintainers. Because nothing was removed, no removal search of `paper/sections/*.tex` or `docs/literature_review.md` was needed.

Two entries are closest to the line. If the owner reads the rule more strictly, these are the candidates, with where they are used (22:48 UTC) and a replacement already in the bib:

Key	Why borderline	Used in	Suggested replacement
costarelli2024gamebench	Peer-reviewed but non-archival NeurIPS 2024 workshop paper by a multi-university student team (Olin, UT Dallas, Columbia, Oxford, UCL, ShanghaiTech, Manchester, two independent), not an established lab	<code>docs/literature_review.md</code> §2.1 (annotation, original line 79); not cited in <code>paper/sections/*.tex</code> at audit time	duan2024gtbench (NeurIPS 2024 main track game-theoretic benchmark)
gao2023lmeval	Software release on Zenodo (DOI, maintainers' recommended citation), not a paper or model documentation	<code>docs/literature_review.md</code> §2.7 (annotation, original line 414); not cited in <code>paper/sections/*.tex</code> at audit time	biderman2024lessons (EleutherAI paper describing lm-evaluation-harness)

(d) Entries kept with residual doubt

- robinson2023mcsb: the ICLR 2023 record (iclr.cc poster page, OpenReview yKbprarjc5B) lists Joshua Robinson and David Wingate. arXiv:2210.12353 also lists Christopher Michael Rytting. The bib follows the venue record. If the camera-ready PDF on OpenReview shows three authors, add Rytting back as the second author.
- shinn2023reflexion: the NeurIPS 2023 proceedings page lists five authors. arXiv v4 (10 Oct 2023) adds Edward Berman. The bib follows the proceedings.
- hoffmann2022chinchilla, duan2024gtbench, dettmers2022lmlnt8: proceedings index title and printed paper title differ (see Method). For Hoffmann et al., the NeurIPS index also orders Vinyals before Rae and uses the given names Thomas, Katherine and Jack, while arXiv has Rae before Vinyals. The bib follows the index for both title and authors, and gives the printed title in note.
- towers2025gymnasium: the arXiv record, the Farama README and (per search) the proceedings PDF put Jordan Terry third, but the NeurIPS virtual-site metadata lists J. K. Terry later. The paper order is kept.
- jiwatode2026spatial: arXiv-only (22 July 2026). The authors' arXiv comment says "To be published at COG 2026" (IEEE CoG, Madrid, 1-4 September 2026), but no proceedings or IEEE Xplore record was found yet. Replace with the CoG record when it appears. The review's Section 5 calls it "(CoG

2026)". That prose was not edited.

- `dilkes2025reinforced`: a single-version arXiv preprint (14 August 2025) from the University of Southampton. It is acceptable as an academic-lab preprint, but it is not peer reviewed and the review leans on it for H3 (the 3B > 72B Frozen Lake result).
- `costarelli2024gamebench` and `gao2023lmeval`: see the Removed section above.
- `zheng2025qwen3quant`: the journal version (Visual Intelligence 4, article 11, 16 April 2026; DOI confirmed on Springer) adds Zining Wang. The author order was taken from the authors' repository citation, because the Springer author block was not readable here.
- `olmo2025olmo2`: upgraded to the shorter COLM 2025 version ("2 OLMo 2 Furious (COLM's Version)", first author Evan Pete Walsh). Its abstract covers the 7B/13B/32B family. For details that exist only in the longer arXiv v3 report (8 October 2025), cite arXiv:2501.00656, which is named in the entry's note.
- `pomerleau1988alvin`: listed by NeurIPS under NIPS 1988; the Morgan Kaufmann volume appeared in 1989 (CMU Robotics Institute record, which also gives pp. 305-313). The year is kept as 1988 to match the NeurIPS listing, with a note.
- `google2026gemma4`: the official announcement is cited. A Gemma 4 Technical Report now exists (arXiv:2607.02770). Consider adding it if the paper cites Gemma 4 architecture details.
- `qwen2025qwen3github`, `qwen2026github`, `meta2025llamamodels`, `hf2025smollm3`, `google2025gemma3n`, `google2026gemma4`, `mistral2025*`: first-party web documentation, kept under the "official model documentation" rule. These pages can change, so access dates are recorded.
- `efron1993bootstrap`: the original imprint is New York: Chapman & Hall, 1993 (WorldCat/Open Library, LCCN 93004489). The Taylor & Francis DOI belongs to the publisher's current listing, which dates the book 1994.

Consequences for the paper and the review

- **In-text labels produced by agsm.** When two entries share an "X et al." label and year but have different author lists, `agsm.bst` disables "et al." for those entries and prints every surname (its "confusion" pass). It adds a/b suffixes only when the full surname lists are identical. With the current bib this affects `wei2022cot` / `wei2022emergent`, `zhou2023lima` / `zhou2023ifeval`, `lin2024urial` / `lin2024awq`, `wang2024myanswerc` / `wang2024lookattext`, and `liu2025mindstep` / `liu2025rational` / `liu2025quanthurts`, which will cite as, e.g., "Wei, Wang, Schuurmans, Bosma, Ichter, Xia, Chi, Le and Zhou (2022)". The three Mistral AI 2025 pages become 2025a/b/c (`mistral2025mistral3`, `mistral2025small3`, `mistral2025small31`). This is agsm's behaviour, not a bib error; a different `.bst` would be needed for "et al." with year suffixes.
- **Citations whose rendered form changed.** `alain2016probes` now renders as 2017; `zheng2025qwen3quant` as Zheng et al. (2026); `deepseekai2025r1` as Guo et al. (2025); `olmo2025olmo2` as Walsh et al. (2025); `qwen2024qwen25` as Yang et al. (2024); `hf2025smollm3` as Bakouch et al. (2025); `robinson2023mcsb` as Robinson and Wingate (2023). `\citet`/`\citep` update automatically, but check any hand-written author names next to these keys. At audit time `paper/sections/setup.tex` cites `qwen2024qwen25`, `olmo2025olmo2` and `deepseekai2025r1`, and `benchmark.tex`, `introduction.tex` and `related_work.tex` cite `robinson2023mcsb`.
- **docs/literature_review.md (mechanical edit only).** A script replaced the 134 bracketed key groups in the prose with parenthetical Harvard citations using exactly the agsm labels above, keeping the order within each group and merging same-author runs as natbib does. It then appended a "Reference list" of the 106 cited works, generated from the audited bib and sorted in agsm order. A check confirmed that the text outside the 134 citation groups is byte-identical to the original, and re-running the script is a no-op. The ``[key]`` labels after each annotation header are inline code (identifiers linking each annotation to `refs.bib`), so they were left as they are.
- **Review prose now out of date (not edited, per instructions).** The opening note still says "106 verified BibTeX entries" (now 113). In Section 4, the "Venues not confirmed, so cited as arXiv" list is out of date: "Look at the Text" (COLM 2024), PAPRIKA (ICML 2025), "Quantization Hurts Reasoning?" (COLM 2025), Alain and Bengio (ICLR 2017 Workshop Track) and GameBench (NeurIPS 2024 workshop) now cite published versions, and SmoLM2 stays arXiv. The "Author lists only partly confirmed" list is resolved (see table). The Reflexion line no longer matches the bib, which follows the NeurIPS author list. The Gemma 4 technical report now exists (arXiv:2607.02770), and its Apache-2.0 licence is confirmed by Google's announcement. OLMo 2 now cites the COLM 2025 version, and the Qwen3 quantisation study cites its 2026 Visual Intelligence version.

Added after the audit: decision models

Nine entries on "System One" decision models (TypeSafe's Jev and its open-weight relatives) were added on 8 October 2026 for the comparison section. They are official announcements, documentation and model cards of the releasing organisations (TypeSafe AI, Cloudflare, Perplexity, the Kev repository) plus TechCrunch and The New Stack reporting. Kev's README was read directly; the TypeSafe, Cloudflare, Perplexity and TechCrunch pages were confirmed through search-engine renderings of the pages themselves because direct fetches were blocked from the writing environment. TechCrunch bylines are not recorded and should be added when the pages are reachable: Harvard style prefers the journalist's name over the publication's.

key	source type	verified via
<code>typesafe2026jev</code>	official announcement	<code>typesafe.ai</code> blog (search rendering)
<code>typesafe2026choice</code>	official documentation	<code>docs.typesafe.ai</code> (search rendering)
<code>typesafe2026systemone</code>	official documentation	<code>docs.typesafe.ai</code> (search rendering)
<code>techcrunch2026jev</code>	reporting	<code>techcrunch.com</code> (search rendering, dated 18 Sep 2026)
<code>techcrunch2026decisionmodels</code>	reporting	<code>techcrunch.com</code> (search rendering, dated 1 Oct 2026)
<code>palmer2026kev</code>	software + model cards	<code>github.com/jaredpalmer/kev</code> README (fetched)
<code>cloudflare2026clef</code>	official announcement	<code>blog.cloudflare.com</code> and <code>developers.cloudflare.com</code> changelog (search rendering)
<code>perplexity2026decider</code>	model card + docs	<code>huggingface.co/perplexity-ai</code> (search rendering)
<code>thenewstack2026jev</code>	reporting	<code>thenewstack.io</code> (search rendering)
<code>convai2026laya</code>	model cards + official site	<code>huggingface.co/convaiinnovations</code> and <code>laya.convaiinnovations.com</code> (search rendering)
<code>lee2026semif</code>	software repository	<code>github.com/TheoLeeCJ/SemIf-OpenJev</code> (search rendering of the README)

Two arXiv preprints that evaluate decision models (arXiv:2609.24574 on text annotation; arXiv:2609.28940 on penetration-testing harnesses, single independent author) were found but not cited: the first's authorship could not be confirmed and the second is not from an established lab, so neither meets the source rule.